

LECTURE 11: EFFICIENT ESTIMATORS FOR AVERAGE TREATMENT EFFECTS UNDER UNCONFOUNDEDNESS

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OUTLINE

1. Three Infeasible Estimators for Average Treatment Effects
2. Three Feasible and Efficient Estimators for Average Treatment Effects
3. Double Robustness
4. Policy Learning

SET UP

- Treatment indicator: $W_i \in \{0, 1\}$, Potential Outcomes $Y_i(C), Y_i(T)$
- Covariate/Pre-treatment Variables X_i , not affected by treatment.
- Realized/Observed Outcome: $Y_i^{\text{obs}} \equiv W_i \cdot Y_i(T) + (1 - W_i) \cdot Y_i(C)$.
- Estimand, population average treatment effect

$$\tau \equiv \mathbb{E} \left[Y_i(T) - Y_i(C) \right]$$

- Assumption 1: unconfoundedness

$$W_i \perp\!\!\!\perp \left(Y_i(C), Y_i(T) \right) \mid X_i$$

- Assumption 2: Overlap

$$0 < \text{pr}(W_i = 1 | X_i) < 1$$

- combined: ignorable treatment assignment
- Assumption 3: Random sample from large population

SEMI-PARAMETRIC VARIANCE BOUND

- Variance bound:

$$\sqrt{N}(\hat{\tau} - \tau) \xrightarrow{d} \mathcal{N}(0, \mathbb{V}),$$

- And:

$$\mathbb{V} \geq \mathbb{V}^{\text{spvb}} \equiv \mathbb{E} \left[\frac{\sigma_T^2(X_i)}{e(X_i)} + \frac{\sigma_C^2(X_i)}{1 - e(X_i)} + (\tau(X_i) - \tau)^2 \right]$$

- The **influence function** is

$$\varphi(w, x, y) \equiv w \frac{y - \mu_T(x)}{e(x)} - (1 - w) \frac{y - \mu_C(x)}{1 - e(x)} + (\mu_T(x) - \mu_C(x)) - \tau$$

- so that

$$\hat{\tau} = \tau + \frac{1}{N} \sum_{i=1}^N \varphi(W_i, X_i, Y_i) + o_p(N^{-1/2}), \quad \mathbb{V}^{\text{spvb}} = \mathbb{E} \left[\varphi(W_i, X_i, Y_i)^2 \right]$$

IMPLICATION

- Suppose there exists an estimator $\hat{\tau}$ that satisfies

$$\hat{\tau} = \tau + \frac{1}{N} \sum_{i=1}^N \varphi(Y_i, W_i, X_i) + o_p \left(N^{-1/2} \right)$$

- Then:

$$\sqrt{N} (\hat{\tau} - \tau) = \frac{1}{\sqrt{N}} \sum_{i=1}^N \varphi(Y_i, W_i, X_i) + o_p(1)$$

- $\implies$ that estimator achieves the semi-parametric efficiency bound.
- Such an estimator would be asymptotically linear and semi-parametrically efficient.
- It turns out there are a bunch of estimators that satisfy this condition.
- What do they look like, and how do we choose between them?

THREE INFEASIBLE ESTIMATORS

- Infeasible Influence Function Estimator

$$\tilde{\tau}^{\text{inf}} = \frac{1}{N} \sum_{i=1}^N \left\{ \left(\mu_T(X_i) - \mu_C(X_i) \right) + \left(W_i \frac{Y_i - \mu_T(X_i)}{e(X_i)} - (1 - W_i) \frac{Y_i - \mu_C(X_i)}{1 - e(X_i)} \right) \right\}$$

- Infeasible Regression estimator

$$\tilde{\tau}^{\text{regr}} = \frac{1}{N} \sum_{i=1}^N \left\{ \mu_T(X_i) - \mu_C(X_i) \right\}$$

- Infeasible Inverse Propensity Score Weighting estimator

$$\tilde{\tau}^{\text{ipw}} = \frac{1}{N} \sum_{i=1}^N \left\{ W_i \frac{Y_i}{e(X_i)} - (1 - W_i) \frac{Y_i}{1 - e(X_i)} \right\}$$

- Then:

$$\mathbb{V} \left(\sqrt{N} (\tilde{\tau}^{\text{regr}} - \tau) \right) \leq \underbrace{\mathbb{V} \left(\sqrt{N} (\tilde{\tau}^{\text{inf}} - \tau) \right)}_{=\mathbb{V}_{\text{spvb}}} \leq \mathbb{V} \left(\sqrt{N} (\tilde{\tau}^{\text{ipw}} - \tau) \right)$$

A SIMPLE EXAMPLE

- Suppose

$$Y_i(\text{C}) = \alpha_{\text{C}} + X_i\beta_{\text{C}} + \varepsilon_i, \quad Y_i(1) = \alpha_{\text{T}} + X_i\beta_{\text{T}} + \eta_i$$

- disturbances:

$$\eta_i = \varepsilon_i \sim \mathcal{N}(0, \sigma^2), \quad \sigma^2 > 0$$

- covariates

$$X_i \sim \mathcal{N}(0, \sigma_X^2)$$

- randomized assignment

$$W_i \sim \mathcal{B}(0, 1/2)$$

A SIMPLE EXAMPLE

- Influence function is

$$\varphi(w, x, y) \equiv (4w - 2)\varepsilon + \left(\alpha_T - \alpha_C + x(\beta_T - \beta_C) \right) - \tau$$

- Then **infeasible influence function estimator** is efficient:

$$\mathbb{V} \left(\sqrt{N} \left(\tilde{\tau}^{\text{infl}} - \tau \right) \right) = \mathbb{V}^{\text{spvb}} = 4\sigma^2 + (\beta_T - \beta_C)^2 \sigma_X^2$$

- **Infeasible regression estimator** has lower variance than variance bound:

$$\mathbb{V} \left(\sqrt{N} \left(\tilde{\tau}^{\text{regr}} - \tau \right) \right) = (\beta_T - \beta_C)^2 \sigma_X^2 < \mathbb{V}^{\text{spvb}}$$

- **Infeasible ipw estimator** has larger variance than variance bound:

$$\mathbb{V} \left(\sqrt{N} \left(\tilde{\tau}^{\text{ipw}} - \tau \right) \right) = 4\sigma^2 + 2(\beta_T^2 + \beta_C^2) \sigma_X^2 \geq \mathbb{V}^{\text{spvb}}$$

FEASIBLE VERSIONS WITH ESTIMATED FUNCTIONS

- Influence function estimator

$$\hat{\tau}^{\text{inf}} = \frac{1}{N} \sum_{i=1}^N \left\{ \left(\hat{\mu}_{\text{T}}(X_i) - \hat{\mu}_{\text{C}}(X_i) \right) + \left(W_i \frac{Y_i - \hat{\mu}_{\text{T}}(X_i)}{\hat{e}(X_i)} - (1 - W_i) \frac{Y_i - \hat{\mu}_{\text{C}}(X_i)}{1 - \hat{e}(X_i)} \right) \right\}$$

- Regression estimator

$$\hat{\tau}^{\text{regr}} = \frac{1}{N} \sum_{i=1}^N \left\{ \hat{\mu}_{\text{T}}(X_i) - \hat{\mu}_{\text{C}}(X_i) \right\}$$

- Inverse Propensity Score Weighting estimator

$$\hat{\tau}^{\text{ipw}} = \frac{1}{N} \sum_{i=1}^N \left\{ \left(W_i \frac{Y_i}{\hat{e}(X_i)} - (1 - W_i) \frac{Y_i}{1 - \hat{e}(X_i)} \right) \right\}$$

- Under some regularity conditions, **all are first-order efficient(!)**:

$$\sqrt{N}(\hat{\tau}^{\text{inf}} - \tau) \sim \sqrt{N}(\hat{\tau}^{\text{regr}} - \tau) \sim \sqrt{N}(\hat{\tau}^{\text{ipw}} - \tau) \sim \mathcal{N}(0, \mathbb{V}^{\text{spvb}})$$

EXAMPLE TO GIVE SOME INTUITION FOR THIS RESULT

- Single binary regressor: $X_i \in \{0, 1\}$

$$\mu_W(x) = \mu_{WX}, \quad \hat{\mu}_W(x) = \bar{Y}_{WX} \quad e(x) = p_x, \quad \hat{e}(x) = \bar{W}_x$$

- Infeasible regression estimator:

$$\hat{\tau}^{\text{regr}} = \bar{X} \cdot (\mu_T(1) - \mu_C(1)) + (1 - \bar{X}) \cdot (\mu_T(0) - \mu_C(0))$$

- Feasible regression estimator:

$$\hat{\tau}^{\text{regr}} = \bar{X} \cdot (\bar{Y}_{T1} - \bar{Y}_{C1}) + (1 - \bar{X}) \cdot (\bar{Y}_{T0} - \bar{Y}_{C0})$$

IPW ESTIMATOR

- weighting estimator

$$\begin{aligned}\hat{\tau}^{\text{ipw}} &= \frac{1}{N} \sum_{i=1}^N \left\{ \frac{Y_i W_i}{\hat{e}(X_i)} - \frac{Y_i(1 - W_i)}{1 - \hat{e}(X_i)} \right\} \\&= \frac{1}{N} \sum_{i=1}^N \left\{ \frac{X_i Y_i W_i}{\hat{e}(1)} + \frac{(1 - X_i) Y_i W_i}{\hat{e}(0)} - \frac{X_i Y_i (1 - W_i)}{1 - \hat{e}(1)} - \frac{(1 - X_i) Y_i (1 - W_i)}{1 - \hat{e}(0)} \right\} \\&= \frac{1}{N} \sum_{i=1}^N \left\{ \frac{X_i \bar{Y}_{T1} W_i}{\bar{W}_1} + \frac{(1 - X_i) \bar{Y}_{T0} W_i}{\bar{W}_0} - \frac{X_i \bar{Y}_{C1} (1 - W_i)}{1 - \bar{W}_1} - \frac{(1 - X_i) \bar{Y}_{C0} (1 - W_i)}{1 - \bar{W}_0} \right\} \\&= \frac{1}{N} \sum_{i=1}^N \left\{ \frac{X_i \bar{Y}_{T1} \bar{W}_1}{\bar{W}_1} + \frac{(1 - X_i) \bar{Y}_{T0} \bar{W}_0}{\bar{W}_0} - \frac{X_i \bar{Y}_{C1} (1 - \bar{W}_1)}{1 - \bar{W}_1} - \frac{(1 - X_i) \bar{Y}_{C0} (1 - \bar{W}_0)}{1 - \bar{W}_0} \right\} \\&= \frac{1}{N} \sum_{i=1}^N \{ X_i \bar{Y}_{T1} + (1 - X_i) \bar{Y}_{T0} - X_i \bar{Y}_{C1} - (1 - X_i) \bar{Y}_{C0} \} \\&= \bar{X}(\bar{Y}_{T1} - \bar{Y}_{C1}) + (1 - \bar{X})(\bar{Y}_{T0} - \bar{Y}_{C0}) = \hat{\tau}^{\text{regr}}\end{aligned}$$

INFLUENCE FUNCTION ESTIMATOR

- influence function estimator

$$\begin{aligned}\hat{\tau}^{\text{inf}} &= \frac{1}{N} \sum_{i=1}^N \left\{ \left(\hat{\mu}_T(X_i) - \hat{\mu}_C(X_i) \right) + \left(w_i \frac{Y_i - \hat{\mu}_T(X_i)}{\hat{e}(X_i)} - (1 - w_i) \frac{Y_i - \hat{\mu}_C(X_i)}{1 - \hat{e}(X_i)} \right) \right\} \\ &= \hat{\tau}^{\text{regr}} + \hat{\tau}^{\text{ipw}} - \frac{1}{N} \sum_{i=1}^N \left\{ \left(w_i \frac{\hat{\mu}_T(X_i)}{\hat{e}(X_i)} - (1 - w_i) \frac{\hat{\mu}_C(X_i)}{1 - \hat{e}(X_i)} \right) \right\}\end{aligned}$$

- Focusing on the last term:

$$\begin{aligned}&\frac{1}{N} \sum_{i=1}^N \left\{ w_i \frac{\hat{\mu}_T(X_i)}{\hat{e}(X_i)} - (1 - w_i) \frac{\hat{\mu}_C(X_i)}{1 - \hat{e}(X_i)} \right\} \\ &= \frac{1}{N} \sum_{i=1}^N \{ X_i \bar{Y}_{T1} + (1 - X_i) \bar{Y}_{T0} - X_i \bar{Y}_{C1} - (1 - X_i) \bar{Y}_{C0} \} = \hat{\tau}^{\text{regr}}\end{aligned}$$

- Hence

$$\hat{\tau}^{\text{infl}} = \hat{\tau}^{\text{regr}} + \hat{\tau}^{\text{ipw}} - \hat{\tau}^{\text{regr}} = \hat{\tau}^{\text{regr}}$$

- And, they are all **identical!**

HOW TO CHOOSE BETWEEN $\hat{\tau}^{\text{reg}}$, $\hat{\tau}^{\text{ipw}}$, AND $\hat{\tau}^{\text{inf}}$

- The regression estimator relies on accurate estimation of the conditional expectations $\mu_w(\cdot)$
- The IPW estimator relies on accurate estimation of the propensity score $e(\cdot)$
- The error of the influence function estimator depends on the **product** of the errors $(\hat{\mu}_w(x) - \mu_w(x))(\hat{e}(x) - e(x))$, and so is less sensitive to errors in either.
- Regression and IPW estimators require more restrictions on smoothness of $\mu(\cdot)$ or $e(\cdot)$ than the influence function estimator.

DOUBLE ROBUSTNESS OF INFLUENCE FUNCTION ESTIMATOR

- Consider a regression model:

$$Y_i = \alpha + \tau W_i + X_i^\top \beta + \varepsilon$$

- the propensity score

$$e(x) = \text{pr}(W_i = 1 | X_i = x)$$

- Consider the weighted least squares estimator

$$\min_{\alpha, \beta, \tau} \sum_{i=1}^N \left(Y_i - \alpha + \tau W_i + X_i^\top \beta \right)^2 \omega(W_i, X_i),$$

- weights

$$\omega(w, x) = \frac{w}{e(x)} + \frac{1-w}{1-e(x)} = \frac{1}{e(x)^w (1-e(x))^{1-w}}$$

DOUBLE ROBUSTNESS OF

- Weighted least squares estimator is

$$\begin{pmatrix} \hat{\alpha} \\ \hat{\tau} \\ \hat{\beta} \end{pmatrix} = \left(\sum_{i=1}^N \omega(W_i, X_i) \begin{pmatrix} 1 \\ W_i \\ X_i \end{pmatrix} \begin{pmatrix} 1 \\ W_i \\ X_i \end{pmatrix}^{\top} \right)^{-1} \left(\sum_{i=1}^N \omega(W_i, X_i) \begin{pmatrix} 1 \\ W_i \\ X_i \end{pmatrix} Y_i \right)$$

DOUBLE ROBUSTNESS

- Suppose that the regression model is **correctly specified**:

$$\mathbb{E}[Y_i|X_i, W_i] = \alpha + \tau W_i + X_i' \beta$$

- expected value of weighted least squares estimator **given** (X_i, W_i) , for $i = 1, \dots, N$ is

$$\left(\sum_{i=1}^N \omega(W_i, X_i) \begin{pmatrix} 1 \\ W_i \\ X_i \end{pmatrix} \begin{pmatrix} 1 \\ W_i \\ X_i \end{pmatrix}^\top \right)^{-1} \left(\sum_{i=1}^N \omega(W_i, X_i) \begin{pmatrix} 1 \\ W_i \\ X_i \end{pmatrix} (\alpha + \tau W_i + X_i' \beta) \right)$$
$$= \begin{pmatrix} \alpha \\ \tau \\ \beta \end{pmatrix}$$

- This holds **irrespective** of form of $\omega(w, x)$. Hence the weights need not be right if the conditional expectation is correctly specified.

DOUBLE ROBUSTNESS

- Same holds the other way.
- key is that in the weighted sample W_i and X_i are independent:

$$\mathbb{E}[h(X_i)(W_i - 1/2)\omega(W_i, X_i)] = 0 \quad \forall h(\cdot)$$

- This follows because

$$\mathbb{E}[\omega(W_i, X_i)] = 2, \quad \mathbb{E}[W_i\omega(W_i, X_i)] = 1,$$

and

$$\begin{aligned} & \mathbb{E}[h(X_i)(W_i - 1/2)\omega(W_i, X_i)|X_i] \\ &= \mathbb{E}[h(X_i)W_i\omega(W_i, X_i)|X_i] - (1/2)\mathbb{E}[h(X_i)\omega(W_i, X_i)|X_i] \\ &= \mathbb{E}\left[h(X_i)W_i\left(\frac{W_i}{e(X_i)} + \frac{1 - W_i}{1 - e(X_i)}\right)\middle|X_i\right] - (1/2)\mathbb{E}\left[h(X_i)\left(\frac{W_i}{e(X_i)} + \frac{1 - W_i}{1 - e(X_i)}\right)\middle|X_i\right] \\ &= \mathbb{E}\left[h(X_i)\frac{W_i}{e(X_i)}\middle|X_i\right] - (1/2)\mathbb{E}\left[h(X_i)\left(\frac{W_i}{e(X_i)} + \frac{1 - W_i}{1 - e(X_i)}\right)\middle|X_i\right] \\ &= \mathbb{E}[h(X_i)] - \mathbb{E}[h(X_i)] = 0 \end{aligned}$$

DOUBLE ROBUSTNESS OF INFLUENCE FUNCTION ESTIMATOR

- First

$$\varphi(y, w, x) \equiv \left(\mu_T(x) - \mu_C(x) \right) + w \frac{y - \mu_T(x)}{e(x)} - (1 - w) \frac{y - \mu_C(x)}{1 - e(x)} - \tau$$

- So,

$$\hat{\tau} = \frac{1}{N} \sum_{i=1}^N \left\{ \varphi(Y_i, W_i, X_i) + \tau \right\} + o_p(N^{-1/2})$$

$$= \frac{1}{N} \sum_{i=1}^N \left(\mu_T(X_i) - \mu_C(X_i) \right) + \frac{1}{N} \sum_{i=1}^N \left(W_i \frac{Y_i - \mu_T(X_i)}{e(X_i)} - (1 - W_i) \frac{Y_i - \mu_C(X_i)}{1 - e(X_i)} \right) + o_p(N^{-1/2})$$

- First part $\sum_{i=1}^N (\mu_T(X_i) - \mu_C(X_i)) / N$ has expectation τ , second part has expectation zero.
- This holds if $E[Y_i | W_i, X_i] = \mu_{W_i}(X_i)$, irrespective of whether $e(X_i) = \text{pr}(W_i = 1 | X_i)$

DOUBLE ROBUSTNESS OF INFLUENCE FUNCTION ESTIMATOR

- Also:

$$\varphi(y, w, x) = w \frac{y}{e(x)} - (1 - w) \frac{y}{1 - e(x)} + (e(x) - w) \left\{ \frac{\mu_1(x)}{e(x)} + \frac{\mu_0(x)}{1 - e(x)} \right\} - \tau$$

So,

$$\begin{aligned} \hat{\tau} &= \frac{1}{N} \sum_{i=1}^N \left\{ \varphi(Y_i, W_i, X_i) + \tau \right\} + o_p(N^{-1/2}) \\ &= \frac{1}{N} \sum_{i=1}^N \left(W_i \frac{Y_i}{e(X_i)} - (1 - W_i) \frac{Y_i}{1 - e(X_i)} \right) + \frac{1}{N} \sum_{i=1}^N (e(X_i) - W_i) \left\{ \frac{\mu_T(X_i)}{e(X_i)} + \frac{\mu_C(X_i)}{1 - e(X_i)} \right\} + o_p(N^{-1/2}) \end{aligned}$$

- Again, first part $\sum_{i=1}^N (W_i Y_i / e(X_i) - (1 - W_i) Y_i / (1 - e(X_i))) / N$ has expectation τ , second part has expectation zero.
- Again this holds irrespective of whether $\mu_w(x)$ is the conditional expectation of Y_i given $W_i = w, X_i = x$.

CROSS-FITTING / SAMPLE SPLITTING (BICKEL ET AL, 1993 CHERNOZHUKOV ET AL, 2018)

- It is easier to derive formal properties if we separate the samples where we do the averaging and the estimation of the unknown functions (though less clear whether this matters in practice, according to Peter Bickel).
- Split the sample into two (or more) parts, with $J_i \in \{0, 1\}$, with $\hat{\mu}_w^j(x)$, and $\hat{e}^j(x)$ estimated on sample with $J_i = j$ (without using observations with $J_i \neq j$)
- Then:

$$\hat{\tau}^j = \frac{1}{N/2} \sum_{i: J_i=j} \left\{ \left(\hat{\mu}_T^{1-j}(X_i) - \hat{\mu}_C^{1-j}(x) \right) + W_i \frac{Y_i - \hat{\mu}_T^{1-j}(X_i)}{\hat{e}^{1-j}(X_i)} - (1 - W_i) \frac{Y_i - \hat{\mu}_C^{1-j}(X_i)}{1 - \hat{e}^{1-j}(X_i)} \right\}$$

- This is **not** efficient, but can average over sample splits to make it efficient:

$$\hat{\tau} = \frac{1}{2} \left(\hat{\tau}^0 + \hat{\tau}^1 \right)$$

- Can do multiple splits.

POLICY LEARNING

- We are interested in a policy rule:

$$\pi : \mathcal{X} \mapsto \{0, 1\}$$

where we assign units to a treatment as a function of their covariates.

- Here $\pi(x) = 1$ if a unit with covariates x gets assigned to the treatment group.
- In an unrestricted world the ideal rule is

$$\pi(x) = \mathbf{1}\{\mu_T(x) - \mu_C(x) \geq 0\}.$$

- One feasible approach is a plug-in rule: Assign based on estimates of the regression function:

$$\pi(x) = \mathbf{1}\{\hat{\mu}_T(x) - \hat{\mu}_C(x) \geq 0\}.$$

- This has some optimality properties (Hirano and Porter, 2009).

POLICY LEARNING

- The more recent literature (Kitagawa and Tetenov, 2018; Athey and Wager, 2019) expands on this in two directions.
 - It uses better estimators for the conditional average treatment effect (doubly robust estimators)
 - It allows for restrictions on the set of rules: we may not want very erratic rules, but simple ones that partition the covariate space into fairly regular subspaces.
- The rules live in a set of possible rules Π . This may be sets of the form:

$$\pi(x) = \mathbf{1}\{x' \beta > 0\}, \quad \text{or} \quad \pi(x) = \mathbf{1}\{x_1 < c, x_2 < d\}$$

POLICY LEARNING

- First let us consider estimating the average effect of a policy $\pi(\cdot)$ relative to the policy of assigning everyone to the control group:

$$\tau(\pi) \equiv \mathbb{E} [\pi(X_i)(Y_i(T) - Y_i(C))] = \mathbb{E} [\pi(X_i)\tau(X_i)]$$

- Recall the influence function

$$\varphi(y, w, x) = \left(\mu_T(x) - \mu_C(x) \right) + w \frac{y - \mu_T(x)}{e(x)} - (1 - w) \frac{y - \mu_C(x)}{1 - e(x)} - \tau$$

- So that we can estimate the average treatment effect $\tau = \mathbb{E}[Y_i(T) - Y_i(C)]$ as

$$\hat{\tau} = \frac{1}{N} \sum_{i=1}^N \hat{\Gamma}_i, \quad \text{where} \quad \Gamma_i \equiv \varphi(Y_i, X_i, W_i) + \tau.$$

POLICY LEARNING

- Then we can estimate the average effect of a policy as

$$\hat{\tau}(\pi) = \frac{1}{N} \sum_{i=1}^N \pi(X_i) \hat{\Gamma}_i,$$

- Exactly like estimating a weighted average treatment effect with weights $\pi(X_i)$.
- We could then try to maximize this over the set of policies:

$$\hat{\pi} = \arg \max_{\pi \in \Pi} \frac{1}{N} \sum_{i=1}^N \pi(X_i) \hat{\Gamma}_i$$

- To make this work we would do the same type of cross-fitting as before:
estimate $\hat{\Gamma}_i$ on part of the data **not** involving observation i .

POLICY LEARNING

- What set of decision rules Π may be used here? Athey and Wager use trees.
- For a depth 1 tree, the set of rules splits on one covariate, at a particular threshold:

$$\Pi = \left\{ \pi(x_1, \dots, x_k) = \mathbf{1}\{x_m \leq c\} \right\}$$

- For a depth 2 tree, the rule splits first at one covariate, and then in both leaves splits at another variable:

$$\Pi = \left\{ \pi(x_1, \dots, x_k) = \mathbf{1}\{x_m \leq c, x_j \leq d, \text{ or } x_m > c, x_r \leq e\} \right\},$$

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